

SMART INDIA HACKATHON 2026



TULA तुला

A compliance engine for the Legal Metrology (Packaged Commodities) Rules, 2011

PROBLEM STATEMENT ID

26034

PS CATEGORY

Software

PROBLEM STATEMENT TITLE

Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels

THEME

‹ Theme as listed on the portal ›

TEAM ID

‹ Team ID ›

TEAM NAME (REGISTERED ON PORTAL)

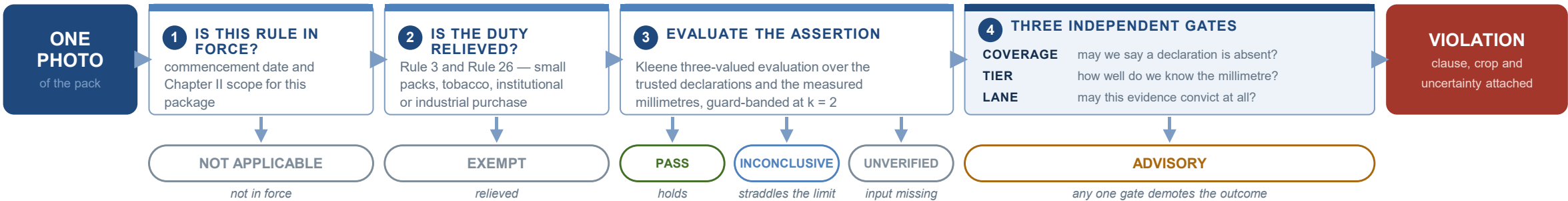
‹ Team name from portal ›

Ministry of Consumer Affairs, Food & Public Distribution · Department of Consumer Affairs



PROPOSED SOLUTION · DETAILED EXPLANATION

Today an officer checks declarations, letter heights and panel area by eye and by ruler. **Tula** takes the photograph through the decision below and returns one of seven verdicts per rule — each carrying its clause, its crop and how far the evidence may be trusted.



A gate can only demote an outcome, never promote it. A rule can fail on the label and still, correctly, not become a violation — because a system that can only say PASS or VIOLATION is forced to guess, and a wrong accusation costs far more than a deferred one.

HOW IT ADDRESSES THE PROBLEM

- **All 18 rules, not a checklist.** Manufacturer, generic name, net quantity, date, MRP, consumer care, unit sale price, country of origin, letter height and panel grouping.
- **Products, images and labels — each on its own footing.** A field capture, a marketplace listing, a citizen's snap and the packer's own artwork enter by different lanes and carry different weight.
- **Ends in an actionable record, not a score.** Clause-cited PDF, editable DOCX and JSON with the crops; the inspector drafts and an independent supervisor approves.

INNOVATION AND UNIQUENESS

- **Seven verdicts, not two.** “I cannot tell from this photograph” becomes a first-class, actionable answer instead of a guess in either direction.
- **No ruler, and no scale needed in order to abstain.** Panel area is bracketed from the photograph itself — the interval held the truth 8 times in 8 — and a size sweep settles the rules whose outcome cannot depend on scale.
- **Rules are dated data, not code.** Reading the Gazette ourselves withdrew a bilingual rule the brief had asked for and corrected a letter height from 1.5 mm to 2.0 mm.

TECHNOLOGIES TO BE USED · METHODOLOGY AND PROCESS FOR IMPLEMENTATION



EVIDENCE STORE

written by every layer above

- SQLite (WAL) · 26 tables
- SHA-256 on every original and working copy
- Append-only audit of who decided what
- Rule-pack archive and rollback
- Missing or altered evidence blocks the report

WORKING PROTOTYPE

running, not a mock-up

- Local console runs capture → review → decide → approve → export
- 1,556 automated tests pass
- Wheel installs clean outside the checkout
- 53-image audit · 40 of 40 checks passed

Inference is CPU-only by design. No GPU anywhere on the serving path, no cloud round trip for a check made on a shop floor — the whole pipeline runs offline on the district-office laptop that is already there.



Your
Team
Name

FEASIBILITY AND VIABILITY



ANALYSIS OF FEASIBILITY · POTENTIAL CHALLENGES AND RISKS · STRATEGIES FOR OVERCOMING THEM

POTENTIAL CHALLENGE	WHY IT IS HARD	OUR STRATEGY FOR OVERCOMING IT	FEASIBILITY EVIDENCE
1 A photograph has no scale	An image records angles, not distances, so a small pack close up and a large one further away give the same picture. Across our 14 real photographs, 0 carried an EXIF subject distance and 12 carried no EXIF.	A printed ArUco card in frame gives a Tier B millimetre. Panel area is bracketed from the image itself — text hull as floor, package edge as ceiling — and reported as an interval, never a point estimate. Where no scale reached the camera, we sweep every package size the declared quantity allows and report only the rules whose outcome cannot change.	8 / 8 the bracketed panel area contained the true value, at 35% median relative uncertainty
2 Real packages defeat OCR	Dot-matrix date codes, glare, curved packs and Devanagari all degrade recognition, and a confident wrong read is far worse than no read.	Every value is scored found vs trusted. An untrusted value is retained as a review candidate and can never on its own establish a violation or an absence. Tesseract with hin data is the second recogniser for Devanagari, and the record says so explicitly when it is missing.	105 / 109 printed values recovered across 24 degraded labels · 0 wrong acceptances on 10 real photographs
3 The Rules keep changing	Clauses commence on different dates and some were later corrected. A hardcoded checker is wrong the day it ships, and silently rewrites past cases when it is upgraded.	Rules are versioned JSON carrying temporal validity and a source citation, not Python. Superseded packs are archived, so a historical record keeps the verdicts it was actually decided under. Reading the Gazette ourselves withdrew one rule outright and corrected two more rather than ship them.	18 rules · 24 sources pack 2026.09.07-legal-review-1, every clause linked to the official PDF it came from
4 A wrong accusation is costly	An automated finding could be mistaken for legal proof, and a marketplace listing image need not even depict the package that ships.	Three independent gates, each able only to demote. Citizen and marketplace lanes are advisory by construction. The inspector drafts, an independent supervisor approves, every action is audited append-only, and no notice is ever issued automatically — the statutory decision stays with a human officer.	1,556 tests full suite green · wheel verified installing clean outside the checkout

STILL OPEN BEFORE DEPLOYMENT independent legal sign-off on the rule pack · physical metrology validation of the printed scale card · a representative multi-state OCR evaluation · formal accessibility certification. We name these rather than claim they are done.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

LEGAL METROLOGY INSPECTOR

field lane · may sustain a violation

- Clause-cited findings, each with the crop it was read from
- No tape measure — panel size comes from the photograph
- Told when a second visit with a card would change nothing

CONTROLLER / SUPERVISOR

approval lane

- Approves or returns findings they did not contribute to
- District-level coverage on the dashboard
- Append-only record of who decided what, on which evidence

MANUFACTURER / PACKER

pre-market lane · Tier A, exact

- Own artwork checked before the print run
- Read from vector source, so no estimation at all
- Fixed on screen rather than after a seizure

CONSUMER / CITIZEN

citizen lane · advisory referral

- A shelf photograph becomes a referral for an inspector
- Deterministic allergen screen, FSS Labelling 2020
- Quotes the exact words it matched, never a guess

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL)

SOCIAL

- One evidence standard in every district — a finding stops depending on which officer holds the ruler
- Hindi in Devanagari read on the same footing as English, because Rule 9(4) puts it there
- Allergen referrals reach the consumer instead of stopping at the file

ECONOMIC

- Fewer wrongful notices — the system abstains instead of guessing
- Fewer wasted second visits: it says outright when a re-capture would change nothing
- Defects caught in artwork, before a run is printed, filled and shipped

ENVIRONMENTAL

- A label defect caught pre-print is packaging that never has to be destroyed or over-stickered
- CPU-only inference adds no GPU procurement
- Runs on the office laptop already in the building

SYSTEM-LEVEL The Jan Vishwas (Amendment of Provisions) Act, 2026 replaced section 36(1) with an improvement notice for a first offence, in force since 1 May 2026. A screen that abstains rather than over-accuses, and leaves the statutory decision with a human officer, is what that regime asks of software.

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

PRIMARY LAW — OFFICIAL GAZETTE PDFs

consumeraffairs.gov.in · sansad.in · fssai.gov.in

- Legal Metrology (Packaged Commodities) Rules, 2011 — the base text
- 2017 amendment and its November 2017 corrigendum — Table I letter heights; the corrigendum sets the first blown/formed cell to 2.0 mm
- G.S.R. 385(E), 2015 — Rule 6(2) consumer care, in force 1 Jan 2016
- G.S.R. 226(E) — prescribed unit-sale-price basis; Rule 6(11) from 1 Jan 2024
- Rajya Sabha AU1181, 11 Feb 2022 — Rule 9(4): Hindi in Devanagari or English
- Jan Vishwas Act 2026, Sch. 66(N) with S.O. 2103(E) — section 36(1), from 1 May 2026
- FSSAI Labelling and Display Regulations compendium — allergen declaration, reg. 5(3)

MEASUREMENT STANDARDS AND METHOD

how a measurement becomes a defensible statement

- ILAC-G8:09/2019 — decision rules and guard bands for statements of conformity. This is why a straddling measurement returns INCONCLUSIVE.
- JCGM 100:2008 (GUM) — expanded uncertainty; every millimetre on this deck is reported at $k = 2$
- Kleene three-valued logic — the basis for a verdict set that is allowed to say 'unknown'
- Garrido-Jurado et al. — ArUco fiducial markers, the in-frame scale reference the scale card prints
- Guard banding applied to Table I of Rule 7(2): the acceptance interval is narrowed by the expanded uncertainty, so the system never convicts inside its own error bar

TECHNOLOGY REFERENCES

everything on the serving path runs on CPU

- PP-OCRv4 detection and recognition, served through RapidOCR on ONNX Runtime
- Tesseract 5 with hin traineddata — Devanagari recognition
- OpenCV-contrib cv2.aruco for markers; zxing-cpp for GTIN decoding
- FastAPI · pydantic v2 · SQLite (WAL) · ReportLab · python-docx
- pillow-heif — HEIC uploads straight from an inspector's phone; originals are never re-encoded
- Open Food Facts was deliberately NOT used: its robots.txt disallows /api for all agents

OUR OWN MEASUREMENTS *every figure on this deck reproduces from the repository, with the script that produced it*

- docs/LEGAL_SOURCE_MATRIX.md — 24 primary sources audited clause by clause; rules withdrawn or corrected where the sources did not bear them out
- docs/OCR_CAPABILITY.md — 105 of 109 printed values recovered across 24 degraded labels
- out/panel-accuracy/results.json — the bracketed panel area contained the truth in 8 of 8
- out/real-label-evaluation-rescan-final/REPORT.md — 12/16 accepted exactly on 10 real photographs, 0 wrong
- data/evaluation/ingredient-panels/ — 7 Wikimedia CC-licensed panels; 0 false positives, both traps clean
- out/hindi-allergen-measurement/results.json — Devanagari screening through real recognition, end to end

Evaluation photographs are Wikimedia Commons CC BY / CC BY-SA; attribution is retained in sources.json and travels with the images. These are unblinded development sets, not independent accuracy estimates — stated as such wherever they are quoted.